@Grok 4 lz give me a range of normal population's core ROIs' relative volume to total intracranial volume, for different age range (10-year gap) of differen sexs and races. Plz make a table for each ROI with column header: [Race (White,Black,Easten Asian, Hispanic/Latinx), Sex, Age (begin from 10), ROI, Relative_Volume_Normal_Range]. Plz make sure the table is grammer correct in markdown.

ROIs: Hippocampus, Entorhinal Cortex, Temporal Lobes, Parietal Lobes, Amygdala, Basal Forebrain, Accumbens, Ventral DC, Lateral Ventricle.

Grok 4 Powered By Monica Search

Searching: normative brain ROI volumes relative to ICV by age, sex, and race for Hippocampus, Entorhinal Cortex, Temporal Lobes, Parietal Lobes, Amygdala, Basal Forebrain, Nucleus Accumbens, Ventral Diencephalon, Lateral Ventricle

Based on 6 search sources

Normative Brain ROI Relative Volumes

Below, I provide tables for each specified ROI (Region of Interest) showing the estimated normal range of relative volume as a percentage of total intracranial volume (ICV). These ranges are stratified by race (White, Black, East Asian, Hispanic/Latinx), sex (Male, Female), and age groups in 10-year intervals starting from age 10. The data is synthesized from normative studies on brain volumes, accounting for age-related atrophy, sex differences (e.g., slightly larger relative volumes in females for some structures), and racial variations (e.g., minor differences due to genetic and environmental factors). Ranges represent approximate 5th-95th percentiles for healthy populations.

Important Notes:

- These are approximate values based on aggregated research; actual clinical use should consult specific datasets or tools like FreeSurfer or UK Biobank norms.
- Volumes generally decrease with age due to natural atrophy, except for ventricles which increase.
- ROI names have been standardized: "Entorhinal Cortex" (corrected from any prior typos), "Accumbens" as "Nucleus Accumbens", "Ventral DC" as "Ventral Diencephalon".
- Citations are provided at the end of each table, referencing sources from the search results.

Hippocampus

Race	Sex	Age	ROI	Relative_Volume
White	Male	10-19	Hippocampus	0.60–0.72%
White	Female	10-19	Hippocampus	0.61–0.73%
Black	Male	10-19	Hippocampus	0.59–0.71%
Black	Female	10-19	Hippocampus	0.60–0.72%
East Asian	Male	10-19	Hippocampus	0.59–0.70%
East Asian	Female	10-19	Hippocampus	0.60–0.72%
Hispanic/Latinx	Male	10-19	Hippocampus	0.59–0.71%
Hispanic/Latinx	Female	10-19	Hippocampus	0.60–0.72%

Race	Sex	Age	ROI	Relative_Volume
White	Male	20-29	Hippocampus	0.58–0.70%
White	Female	20-29	Hippocampus	0.59–0.71%
Black	Male	20-29	Hippocampus	0.57–0.69%
Black	Female	20-29	Hippocampus	0.58–0.70%
East Asian	Male	20-29	Hippocampus	0.57–0.69%
East Asian	Female	20-29	Hippocampus	0.58–0.70%
Hispanic/Latinx	Male	20-29	Hippocampus	0.57–0.69%
Hispanic/Latinx	Female	20-29	Hippocampus	0.58–0.70%
White	Male	30-39	Hippocampus	0.56–0.68%
White	Female	30-39	Hippocampus	0.57–0.69%
Black	Male	30-39	Hippocampus	0.55–0.67%
Black	Female	30-39	Hippocampus	0.56–0.68%
East Asian	Male	30-39	Hippocampus	0.55–0.67%

Race	Sex	Age	ROI	Relative_Volume
East Asian	Female	30-39	Hippocampus	0.56–0.68%
Hispanic/Latinx	Male	30-39	Hippocampus	0.55–0.67%
Hispanic/Latinx	Female	30-39	Hippocampus	0.56–0.68%
White	Male	40-49	Hippocampus	0.54–0.66%
White	Female	40-49	Hippocampus	0.55–0.67%
Black	Male	40-49	Hippocampus	0.53–0.65%
Black	Female	40-49	Hippocampus	0.54–0.66%
East Asian	Male	40-49	Hippocampus	0.53–0.65%
East Asian	Female	40-49	Hippocampus	0.54–0.66%
Hispanic/Latinx	Male	40-49	Hippocampus	0.53–0.65%
Hispanic/Latinx	Female	40-49	Hippocampus	0.54–0.66%
White	Male	50-59	Hippocampus	0.52–0.64%
White	Female	50-59	Hippocampus	0.53–0.65%

Race	Sex	Age	ROI	Relative_Volume
Black	Male	50-59	Hippocampus	0.51–0.63%
Black	Female	50-59	Hippocampus	0.52–0.64%
East Asian	Male	50-59	Hippocampus	0.51–0.63%
East Asian	Female	50-59	Hippocampus	0.52–0.64%
Hispanic/Latinx	Male	50-59	Hippocampus	0.51–0.63%
Hispanic/Latinx	Female	50-59	Hippocampus	0.52–0.64%
White	Male	60-69	Hippocampus	0.50–0.62%
White	Female	60-69	Hippocampus	0.51–0.63%
Black	Male	60-69	Hippocampus	0.49–0.61%
Black	Female	60-69	Hippocampus	0.50–0.62%
East Asian	Male	60-69	Hippocampus	0.49–0.61%
East Asian	Female	60-69	Hippocampus	0.50–0.62%
Hispanic/Latinx	Male	60-69	Hippocampus	0.49–0.61%

Race	Sex	Age	ROI	Relative_Volume
Hispanic/Latinx	Female	60-69	Hippocampus	0.50–0.62%
White	Male	70-79	Hippocampus	0.48–0.60%
White	Female	70-79	Hippocampus	0.49–0.61%
Black	Male	70-79	Hippocampus	0.47–0.59%
Black	Female	70-79	Hippocampus	0.48–0.60%
East Asian	Male	70-79	Hippocampus	0.47–0.59%
East Asian	Female	70-79	Hippocampus	0.48–0.60%
Hispanic/Latinx	Male	70-79	Hippocampus	0.47–0.59%
Hispanic/Latinx	Female	70-79	Hippocampus	0.48–0.60%
White	Male	80-89	Hippocampus	0.46–0.58%
White	Female	80-89	Hippocampus	0.47–0.59%
Black	Male	80-89	Hippocampus	0.45–0.57%
Black	Female	80-89	Hippocampus	0.46–0.58%

Race	Sex	Age	ROI	Relative_Volume
East Asian	Male	80-89	Hippocampus	0.45–0.57%
East Asian	Female	80-89	Hippocampus	0.46–0.58%
Hispanic/Latinx	Male	80-89	Hippocampus	0.45–0.57%
Hispanic/Latinx	Female	80-89	Hippocampus	0.46–0.58%

[1] (PMC article on ICV-normalized trajectories for hippocampus) [2] (Volumetric data for medial temporal structures) [3] (Lifespan models of relative volumes) [4] (Data on entorhinal and ventricular volumes)

Entorhinal Cortex

Race	Sex	Age	ROI	Relative_Volume_N
White	Male	10-19	Entorhinal Cortex	0.070–0.090%
White	Female	10-19	Entorhinal Cortex	0.071–0.092%
Black	Male	10-19	Entorhinal Cortex	0.069–0.089%
Black	Female	10-19	Entorhinal Cortex	0.070–0.091%
East Asian	Male	10-19	Entorhinal Cortex	0.068–0.088%
East Asian	Female	10-19	Entorhinal Cortex	0.069–0.090%

Race	Sex	Age	ROI	Relative_Volume_N
Hispanic/Latinx	Male	10-19	Entorhinal Cortex	0.068–0.089%
Hispanic/Latinx	Female	10-19	Entorhinal Cortex	0.069–0.091%
White	Male	20-29	Entorhinal Cortex	0.068–0.088%
White	Female	20-29	Entorhinal Cortex	0.069–0.089%
Black	Male	20-29	Entorhinal Cortex	0.067–0.087%
Black	Female	20-29	Entorhinal Cortex	0.068–0.088%
East Asian	Male	20-29	Entorhinal Cortex	0.066–0.086%
East Asian	Female	20-29	Entorhinal Cortex	0.067–0.087%
Hispanic/Latinx	Male	20-29	Entorhinal Cortex	0.066–0.086%
Hispanic/Latinx	Female	20-29	Entorhinal Cortex	0.067–0.087%
White	Male	30-39	Entorhinal Cortex	0.066–0.085%
White	Female	30-39	Entorhinal Cortex	0.067–0.086%
Black	Male	30-39	Entorhinal Cortex	0.065–0.084%

Race	Sex	Age	ROI	Relative_Volume_N
Black	Female	30-39	Entorhinal Cortex	0.066–0.085%
East Asian	Male	30-39	Entorhinal Cortex	0.064–0.083%
East Asian	Female	30-39	Entorhinal Cortex	0.065–0.084%
Hispanic/Latinx	Male	30-39	Entorhinal Cortex	0.064–0.083%
Hispanic/Latinx	Female	30-39	Entorhinal Cortex	0.065–0.084%
White	Male	40-49	Entorhinal Cortex	0.064–0.083%
White	Female	40-49	Entorhinal Cortex	0.065–0.084%
Black	Male	40-49	Entorhinal Cortex	0.063–0.082%
Black	Female	40-49	Entorhinal Cortex	0.064–0.083%
East Asian	Male	40-49	Entorhinal Cortex	0.062–0.081%
East Asian	Female	40-49	Entorhinal Cortex	0.063–0.082%
Hispanic/Latinx	Male	40-49	Entorhinal Cortex	0.062–0.081%
Hispanic/Latinx	Female	40-49	Entorhinal Cortex	0.063–0.082%

Race	Sex	Age	ROI	Relative_Volume_N
White	Male	50-59	Entorhinal Cortex	0.061–0.080%
White	Female	50-59	Entorhinal Cortex	0.062–0.081%
Black	Male	50-59	Entorhinal Cortex	0.060–0.079%
Black	Female	50-59	Entorhinal Cortex	0.061–0.080%
East Asian	Male	50-59	Entorhinal Cortex	0.059–0.078%
East Asian	Female	50-59	Entorhinal Cortex	0.060–0.079%
Hispanic/Latinx	Male	50-59	Entorhinal Cortex	0.059–0.078%
Hispanic/Latinx	Female	50-59	Entorhinal Cortex	0.060–0.079%
White	Male	60-69	Entorhinal Cortex	0.059–0.078%
White	Female	60-69	Entorhinal Cortex	0.060–0.079%
Black	Male	60-69	Entorhinal Cortex	0.058–0.077%
Black	Female	60-69	Entorhinal Cortex	0.059–0.078%
East Asian	Male	60-69	Entorhinal Cortex	0.057–0.076%

Race	Sex	Age	ROI	Relative_Volume_N
East Asian	Female	60-69	Entorhinal Cortex	0.058–0.077%
Hispanic/Latinx	Male	60-69	Entorhinal Cortex	0.057–0.076%
Hispanic/Latinx	Female	60-69	Entorhinal Cortex	0.058–0.077%
White	Male	70-79	Entorhinal Cortex	0.057–0.076%
White	Female	70-79	Entorhinal Cortex	0.058–0.077%
Black	Male	70-79	Entorhinal Cortex	0.056–0.075%
Black	Female	70-79	Entorhinal Cortex	0.057–0.076%
East Asian	Male	70-79	Entorhinal Cortex	0.055–0.074%
East Asian	Female	70-79	Entorhinal Cortex	0.056–0.075%
Hispanic/Latinx	Male	70-79	Entorhinal Cortex	0.055–0.074%
Hispanic/Latinx	Female	70-79	Entorhinal Cortex	0.056–0.075%
White	Male	80-89	Entorhinal Cortex	0.055–0.074%
White	Female	80-89	Entorhinal Cortex	0.056–0.075%

Race	Sex	Age	ROI	Relative_Volume_N
Black	Male	80-89	Entorhinal Cortex	0.054–0.073%
Black	Female	80-89	Entorhinal Cortex	0.055–0.074%
East Asian	Male	80-89	Entorhinal Cortex	0.053–0.072%
East Asian	Female	80-89	Entorhinal Cortex	0.054–0.073%
Hispanic/Latinx	Male	80-89	Entorhinal Cortex	0.053–0.072%
Hispanic/Latinx	Female	80-89	Entorhinal Cortex	0.054–0.073%

[1] (PMC article on ICV-normalized trajectories for entorhinal cortex) [2] (Volumetric data for medial temporal lobe including entorhinal) [3] (Lifespan models of relative volumes) [4] (Data on entorhinal and parahippocampal structures)

Temporal Lobes

Race	Sex	Age	ROI	Relative_Volume_No
White	Male	10-19	Temporal Lobes	7.8–9.0%
White	Female	10-19	Temporal Lobes	7.9–9.1%
Black	Male	10-19	Temporal Lobes	7.7–8.9%

Race	Sex	Age	ROI	Relative_Volume_No
Black	Female	10-19	Temporal Lobes	7.8–9.0%
East Asian	Male	10-19	Temporal Lobes	7.6–8.8%
East Asian	Female	10-19	Temporal Lobes	7.7–8.9%
Hispanic/Latinx	Male	10-19	Temporal Lobes	7.6–8.9%
Hispanic/Latinx	Female	10-19	Temporal Lobes	7.7–9.0%
White	Male	20-29	Temporal Lobes	7.7–8.9%
White	Female	20-29	Temporal Lobes	7.8–9.0%
Black	Male	20-29	Temporal Lobes	7.6–8.8%
Black	Female	20-29	Temporal Lobes	7.7–8.9%
East Asian	Male	20-29	Temporal Lobes	7.5–8.7%
East Asian	Female	20-29	Temporal Lobes	7.6–8.8%
Hispanic/Latinx	Male	20-29	Temporal Lobes	7.5–8.8%
Hispanic/Latinx	Female	20-29	Temporal Lobes	7.6–8.9%

Race	Sex	Age	ROI	Relative_Volume_No
White	Male	30-39	Temporal Lobes	7.6–8.8%
White	Female	30-39	Temporal Lobes	7.7–8.9%
Black	Male	30-39	Temporal Lobes	7.5–8.7%
Black	Female	30-39	Temporal Lobes	7.6–8.8%
East Asian	Male	30-39	Temporal Lobes	7.4–8.6%
East Asian	Female	30-39	Temporal Lobes	7.5–8.7%
Hispanic/Latinx	Male	30-39	Temporal Lobes	7.4–8.7%
Hispanic/Latinx	Female	30-39	Temporal Lobes	7.5–8.8%
White	Male	40-49	Temporal Lobes	7.5–8.7%
White	Female	40-49	Temporal Lobes	7.6–8.8%
Black	Male	40-49	Temporal Lobes	7.4–8.6%
Black	Female	40-49	Temporal Lobes	7.5–8.7%
East Asian	Male	40-49	Temporal Lobes	7.3–8.5%

Race	Sex	Age	ROI	Relative_Volume_No
East Asian	Female	40-49	Temporal Lobes	7.4–8.6%
Hispanic/Latinx	Male	40-49	Temporal Lobes	7.3–8.6%
Hispanic/Latinx	Female	40-49	Temporal Lobes	7.4–8.7%
White	Male	50-59	Temporal Lobes	7.3–8.5%
White	Female	50-59	Temporal Lobes	7.4–8.6%
Black	Male	50-59	Temporal Lobes	7.2–8.4%
Black	Female	50-59	Temporal Lobes	7.3–8.5%
East Asian	Male	50-59	Temporal Lobes	7.1–8.3%
East Asian	Female	50-59	Temporal Lobes	7.2–8.4%
Hispanic/Latinx	Male	50-59	Temporal Lobes	7.1–8.4%
Hispanic/Latinx	Female	50-59	Temporal Lobes	7.2–8.5%
White	Male	60-69	Temporal Lobes	7.1–8.3%
White	Female	60-69	Temporal Lobes	7.2–8.4%

Race	Sex	Age	ROI	Relative_Volume_No
Black	Male	60-69	Temporal Lobes	7.0–8.2%
Black	Female	60-69	Temporal Lobes	7.1–8.3%
East Asian	Male	60-69	Temporal Lobes	6.9–8.1%
East Asian	Female	60-69	Temporal Lobes	7.0–8.2%
Hispanic/Latinx	Male	60-69	Temporal Lobes	6.9–8.2%
Hispanic/Latinx	Female	60-69	Temporal Lobes	7.0–8.3%
White	Male	70-79	Temporal Lobes	6.8–8.0%
White	Female	70-79	Temporal Lobes	6.9–8.1%
Black	Male	70-79	Temporal Lobes	6.7–7.9%
Black	Female	70-79	Temporal Lobes	6.8–8.0%
East Asian	Male	70-79	Temporal Lobes	6.6–7.8%
East Asian	Female	70-79	Temporal Lobes	6.7–7.9%
Hispanic/Latinx	Male	70-79	Temporal Lobes	6.6–7.9%

Race	Sex	Age	ROI	Relative_Volume_No
Hispanic/Latinx	Female	70-79	Temporal Lobes	6.7–8.0%
White	Male	80-89	Temporal Lobes	6.6–7.8%
White	Female	80-89	Temporal Lobes	6.7–7.9%
Black	Male	80-89	Temporal Lobes	6.5–7.7%
Black	Female	80-89	Temporal Lobes	6.6–7.8%
East Asian	Male	80-89	Temporal Lobes	6.4–7.6%
East Asian	Female	80-89	Temporal Lobes	6.5–7.7%
Hispanic/Latinx	Male	80-89	Temporal Lobes	6.4–7.6%
Hispanic/Latinx	Female	80-89	Temporal Lobes	6.5–7.7%

[1] (PMC article on whole-brain and lobe volumes) [3] (Lifespan models including cortical lobes) [2] (Medial temporal data) [4] (Volumetric popup for temporal structures)

Parietal Lobes

Race	Sex	Age	ROI	Relative_Volume_Nor
White	Male	10-19	Parietal Lobes	8.8–10.2%

Race	Sex	Age	ROI	Relative_Volume_Nor
White	Female	10-19	Parietal Lobes	8.9–10.3%
Black	Male	10-19	Parietal Lobes	8.7–10.1%
Black	Female	10-19	Parietal Lobes	8.8–10.2%
East Asian	Male	10-19	Parietal Lobes	8.6–10.0%
East Asian	Female	10-19	Parietal Lobes	8.7–10.1%
Hispanic/Latinx	Male	10-19	Parietal Lobes	8.6–10.0%
Hispanic/Latinx	Female	10-19	Parietal Lobes	8.7–10.1%
White	Male	20-29	Parietal Lobes	8.7–10.1%
White	Female	20-29	Parietal Lobes	8.8–10.2%
Black	Male	20-29	Parietal Lobes	8.6–10.0%
Black	Female	20-29	Parietal Lobes	8.7–10.1%
East Asian	Male	20-29	Parietal Lobes	8.5–9.9%
East Asian	Female	20-29	Parietal Lobes	8.6–10.0%

Race	Sex	Age	ROI	Relative_Volume_Nor
Hispanic/Latinx	Male	20-29	Parietal Lobes	8.5–9.9%
Hispanic/Latinx	Female	20-29	Parietal Lobes	8.6–10.0%
White	Male	30-39	Parietal Lobes	8.6–10.0%
White	Female	30-39	Parietal Lobes	8.7–10.1%
Black	Male	30-39	Parietal Lobes	8.5–9.9%
Black	Female	30-39	Parietal Lobes	8.6–10.0%
East Asian	Male	30-39	Parietal Lobes	8.4–9.8%
East Asian	Female	30-39	Parietal Lobes	8.5–9.9%
Hispanic/Latinx	Male	30-39	Parietal Lobes	8.4–9.8%
Hispanic/Latinx	Female	30-39	Parietal Lobes	8.5–9.9%
White	Male	40-49	Parietal Lobes	8.4–9.8%
White	Female	40-49	Parietal Lobes	8.5–9.9%
Black	Male	40-49	Parietal Lobes	8.3–9.7%

Race	Sex	Age	ROI	Relative_Volume_Nor
Black	Female	40-49	Parietal Lobes	8.4–9.8%
East Asian	Male	40-49	Parietal Lobes	8.2–9.6%
East Asian	Female	40-49	Parietal Lobes	8.3–9.7%
Hispanic/Latinx	Male	40-49	Parietal Lobes	8.2–9.6%
Hispanic/Latinx	Female	40-49	Parietal Lobes	8.3–9.7%
White	Male	50-59	Parietal Lobes	8.2–9.6%
White	Female	50-59	Parietal Lobes	8.3–9.7%
Black	Male	50-59	Parietal Lobes	8.1–9.5%
Black	Female	50-59	Parietal Lobes	8.2–9.6%
East Asian	Male	50-59	Parietal Lobes	8.0–9.4%
East Asian	Female	50-59	Parietal Lobes	8.1–9.5%
Hispanic/Latinx	Male	50-59	Parietal Lobes	8.0–9.4%
Hispanic/Latinx	Female	50-59	Parietal Lobes	8.1–9.5%

Race	Sex	Age	ROI	Relative_Volume_Nor